

ShopSphere Architecture Decision Record

Review Service Extraction

The **reviews functionality** was extracted from the ShopSphere monolithic backend into an independently deployed **Review Service**. Reviews were selected because they have their own data, routes, controllers, and MongoDB operations, making them a clearly separable feature. Extracting them allows the review functionality to be developed and deployed independently while the main application continues to handle users, products, carts, and other core functions.

Serverless Integration

The **review statistics calculation** was moved to a **Vercel Serverless Function**. The function calculates review statistics such as the total number of reviews and average rating directly from the review database. Serverless is suitable for this workload because it is a small, independent operation that can execute on demand without requiring a continuously running server. It can therefore run separately from the main application while using resources only when the calculation is performed.

Decision

ShopSphere therefore uses a hybrid architecture: the core application remains as the main backend, reviews operate as an independent REST service, and review-statistics processing runs as a Vercel Serverless Function.